CMU-CS 462: Software Measurement and Analysis

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Earned Value Management

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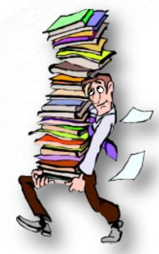


Earned Value Analysis

What Is It ?

Why Do I Need It ?

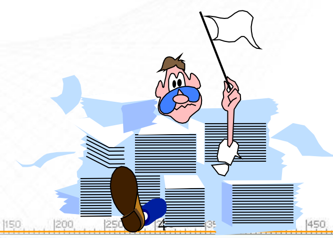
How Do I Do It?

Today's Situation

Need for <u>accurate</u> and <u>consistent</u> status information

Numerous complex (and interrelated) projects

- Projects with many WBS activities
- Virtual offices
- Diverse technology platforms

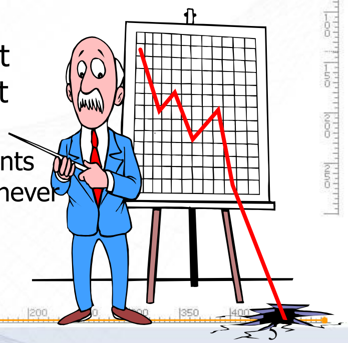


There's Room For Improvement

70% of projects are:
- •Over budget
- •Behind schedule

52% of all projects finish at 189% of their initial budget

And some, after huge investments of time and money, are simply never comple



Source:The Standish Group

How to answer the question: "Have we done what we said we'd do?"

% complete estimating
- % of Budget spent
- % of work done
- % of time elapsed

subjective, incomplete
draws false conclusions

Enter Earned Value Analysis

- "Earned Value Analysis" is:
  - an industry standard way to:
    - measure a project's progress,
    - forecast its completion date and final cost, and
    - provide schedule and budget variances along the way.
- By integrating three measurements, it provides consistent, numerical indicators with which you can evaluate and compare projects.

What's more Important?

Knowing where you are on schedule?

Knowing where you are on budget?

Knowing where you are on work accomplished?

EVA Integrates All Three

It compares the PLANNED amount of work with what has actually been COMPLETED, to determine if *COST* , *SCHEDULE, and WORK ACCOMPLISHED* are progressing as planned.

Work is "Earned" or credited as it is completed.

Earned Value needed because...

Different measures of progress for different types of tasks

Need to "roll up" progress of many tasks into an overall project status

Need for a uniform unit of measure (dollars or work-hours).

Earned Value needed because...

Provides an "Early Warning" signal for prompt corrective action.

Bad news does not age well.

Still time to recover

Timely request for additional funds

And One More Reason

Why You Need EVA

?

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Because You Gotta !

These Set the Stage:

GPRA; 1993

FASA, Title V; 1994

Clinger-Cohen Act; 1996

And Then Along Came OMB! (Circular A-11, Part 7)

"Agencies must use a performance based acquisition management system, based on ANSI/EIA Standard 748, to measure achievement of the cost, schedule, and performance goals."

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OK, So What Is This Stuff?

So, Is This Stuff New ?

It's been around since the sixties.

"Cost/Schedule Control Systems Criteria" (C/SCSC)

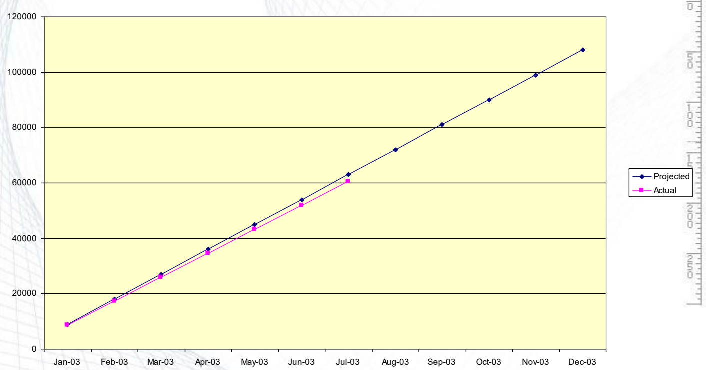
Examples of informal Earned Value Analysis

It's done informally without realizing it.
- •30% time used,
- •30% **$$** spent
- •So, if 30% of the work is done, I must be OK ??

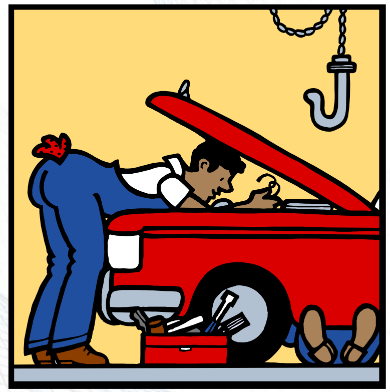
•Shop floor estimates

•Cost comparisons
Budget vs. Actual

How's this project doing?



Let's Take A Look Under The Hood



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But First! - We gotta get organized

- EVA works best when work is 'compartmentalized'.
- Compartmentalization is best achieved with a well-planned Work Breakdown Structure.
- So, how do I create a WBS for a really complex project?

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How am I gonna eat this elephant?

Obviously in small bites.

Proper WBS Design

- One WBS per program
  - Deliverable-oriented
  - Work not in the WBS is out-of-scope
  - Each descending level represents more detail
- Full (and accurate) definition is key
  - Defined deliverable(s)
  - Timeframe for delivery of product
  - Total cost (direct and indirect) to deliver product

Let's Look at an example:

A sample Work Breakdown Structure

- Serve Pizzas to Customers
  - Provide the Place
  - Cook the Food
    - Make the Dough
    - Cook the Sauce
    - Build the Pizza
  - Serve Customers
  - *(Others)*

WBS Units are "Work Packages"

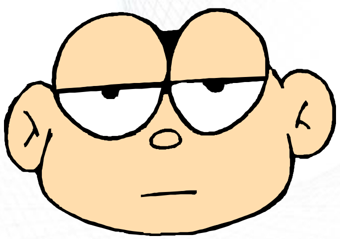
- Lowest level WBS elements
- Have an accompanying narrative
- Have three measurable components
  - Scope of work to be accomplished
  - Total (direct and indirect) cost
  - Timeframe for completion

Control Account Plans

- A CAP is essentially a Work Package with some added features:
  - Assignment of responsibility
    - Organization
    - Individual
  - Division (if necessary) into lower-level Work Packages.
  - Metrics for measuring EV performance
    - Milestones
    - % complete
    - Other

The sum of the CAPs constitutes the Performance Measurement Baseline

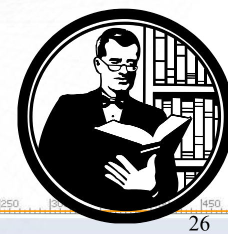
Enough With the WBS Stuff Already !



We came here to talk about Earned Value.

Some New Terms

- BCWS - Budgeted Cost of Work Scheduled
- ACWP - Actual Cost of Work Performed
- BCWP - Budgeted Cost of Work Performed

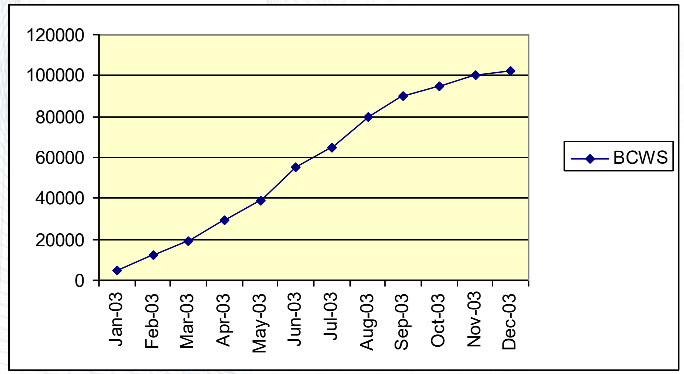


Earned Value Definitions

- BCWS: "Budgeted Cost of Work Scheduled"

Planned cost of the total amount of work scheduled to be performed by the milestone date.

BCWS - Budgeted Cost of Work Scheduled

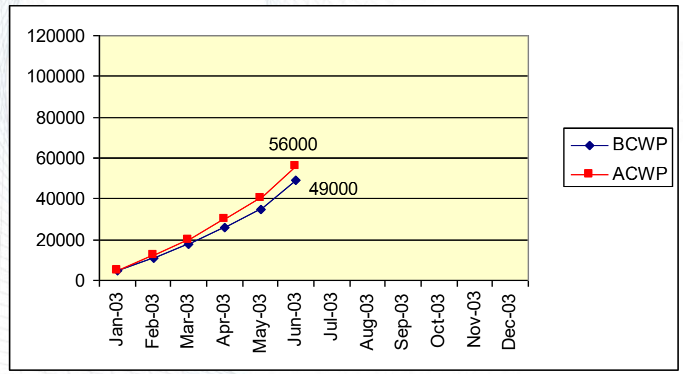


Earned Value Definitions (cont.)

◆ACWP: "Actual Cost of Work Performed"

Cost incurred to accomplish the work that has been done to date.

ACWP - Actual Cost of Work Performed

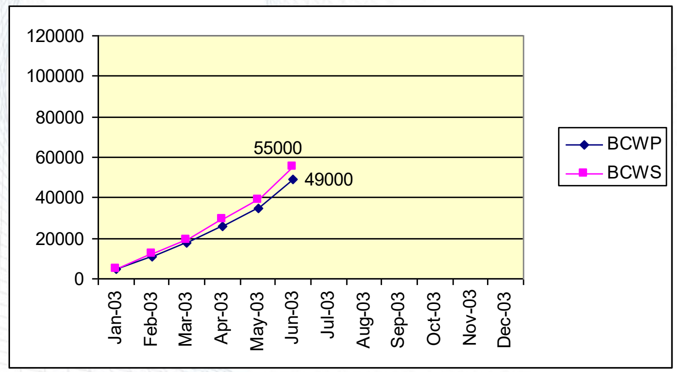


Earned Value Definitions (cont.)

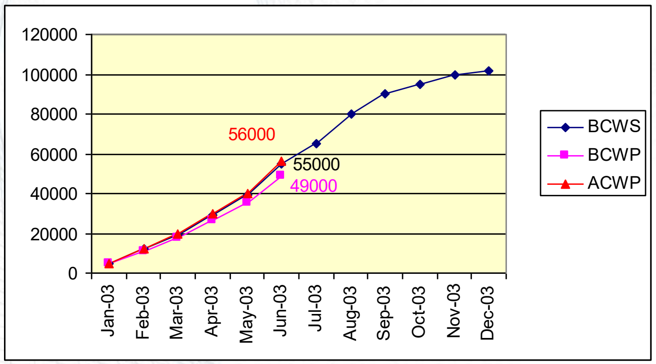
BCWP: Budgeted Cost of Work Performed

The planned (not actual) cost to complete the work that has been done.

BCWP - Budgeted Cost of Work Performed



The Whole Story



Some Derived Metrics

SV: Schedule Variance (BCWP-BCWS)

A comparison of amount of work performed during a given period of time to what was scheduled to be performed.

A negative variance means the project is behind schedule

CV: Cost Variance (BCWP-ACWP)

A comparison of the budgeted cost of work performed with actual cost.

A negative variance means the project is over budget.

Schedule Variance & Cost Variance

Schedule Variance = BCWP-BCWS

$49,000
- 55,000
SV = - $ 6,000

Cost Variance = BCWP-ACWP

$49,000
56,000
CV = - $7,000

Some More Derived Metrics

SPI: Schedule Performance Index

SPI=BCWP/BCWS

SPI<1 means project is behind schedule

CPI: Cost Performance Index CPI= BCWP/ACWP CPI<1 means project is over budget

CSI: Cost Schedule Index (CSI=CPI x SPI)

The further CSI is from 1.0, the less likely project recovery becomes.

Performance Metrics

SPI: BCWP/BCWS

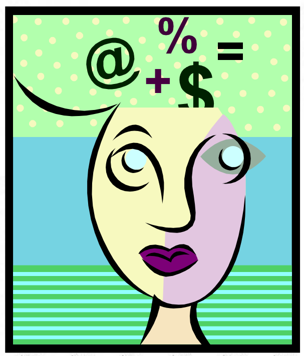
49,000/55,000 = 0.891

CPI: BCWP/ACWP

49,000/56000 = 0.875

CSI: SPI x CPI

.891 x .875 = 0.780



Making Projections

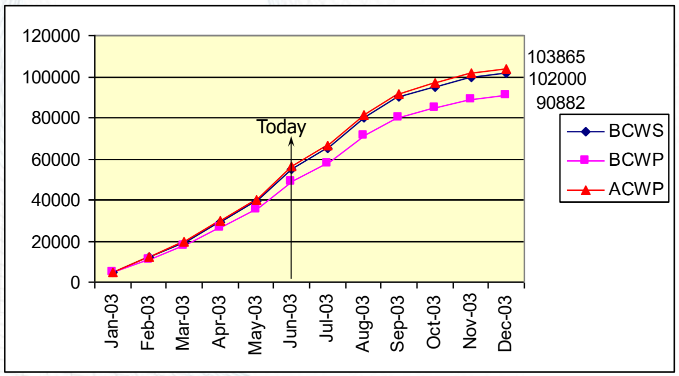
Once a project is 10% complete, the overrun at completion will not be less than the current overrun.

Once a project is 20% complete, the CPI does not vary from its current value by more than 10%.

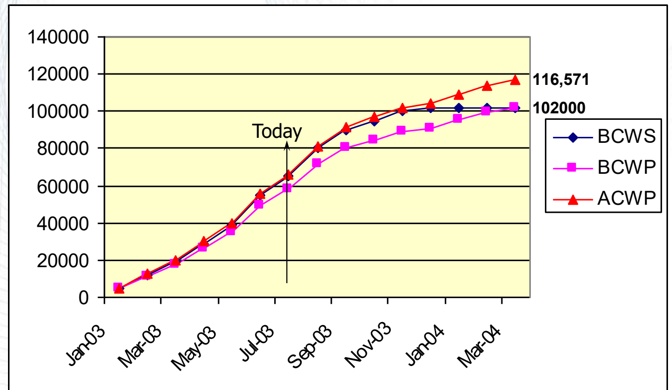
The CPI and SPI are statistically accurate indicators of final cost results.

Source: Defense Acquisition University

Making Projections



Estimate to Complete



A New Criteria

Activities "earn value" as they are completed.

- *The value earned is the WBS budgeted cost of the activity completed to date.*

Value of Earned Value

Schedule Status Reporting
Cost Status Reporting
Forecasting

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But How Do I Do All This Stuff ?

With an Earned Value Management System

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A-11, Part 7 Requires an EVMS

" . . . based on ANSI/EIA Standard 748"

And what does that mean?

➢ANSI/EIA 748 provides a list of guidelines
- Organization
- Planning, Scheduling, and Budgeting
- Accounting Considerations
- Analysis and Management Reports
- Revisions and Data Maintenance

➢But, ANSI/EIA 748 doesn't identify 'approved systems'

A-11, Part 7 Requires an EVMS

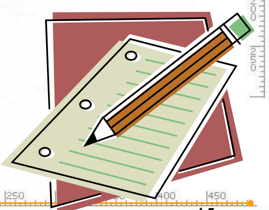
So where do I get one?

➢Buy a prepackaged one. (Lot of 'em around)

➢Make your own.

- Microsoft Project
- Microsoft Excel

Or it could be as simple as this:



Requirements of Earned Value

Proper WBS Design
Baseline Budget Control Accounts
Baseline Schedule
Work measurement by Control Account
work-hours, dollars, units, etc.
Good Project Management Practices

Discuss in group: Shortcomings of Earned Value?

Quantifying/measuring work progress can be difficult.

Time required for data measurement, input, and manipulation can be considerable.

Summary

EVA & EVMS will help reduce guesswork in:

- Measuring performance
- forecasting

Need to get beyond misleading measures of progress.

Reasons to use EVA and EVMS:

- Good project management practice
- OMB requirement

Incorporate into contracts

Earned Value Resources

http://www.pmi.org/

http://www.acq.osd.mil/pm/

ANSI/EIA 748 is available from:

Global Engineering Documents
800-854-7179

Some "Compliant" Systems

Welcom "Cobra"	http://www.welcom.com/
Schedulemaker	http://www.schedulemaker.com/
Planisware "OPX2"	http://www.planisware.com/
RiskTrak	http://www.risktrak.com/index.htm
Winsight	http://www.cs-solutions.com
Primavera Systems	http://www.primavera.com

https://www.youtube.com/watch?v=MO6EOMd-CkA

Earned Value Analysis

Questions/Discussion